Transforming Inputs

Regression models are sensitive to extreme or outlying values in the input space. Inputs with highly skewed or highly kurtotic distributions can be selected over inputs that yield better overall predictions.

To avoid this problem, analysts often regularize the input distributions using a simple transformation. The benefit of this approach is improved model performance. The cost, of course, is increased difficulty in model interpretation.

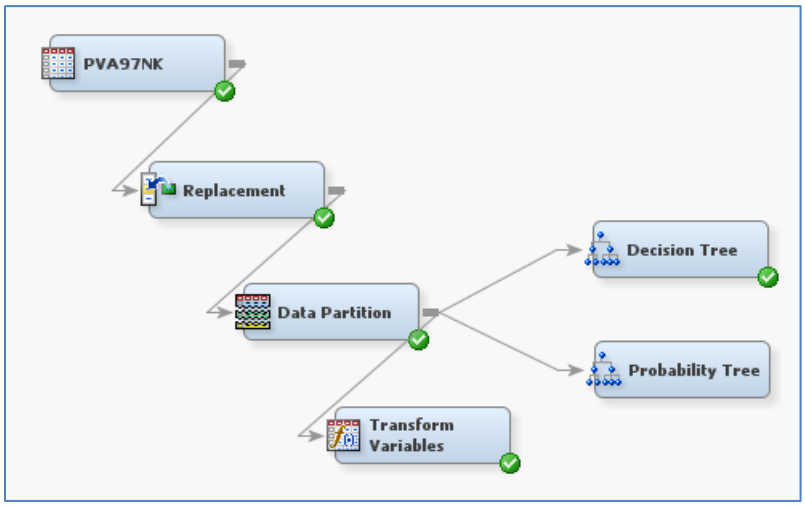
The Transform Variables tool enables you to easily apply standard transformations to a set of inputs.

Use the following steps to transform inputs with the Transform Variables tool:

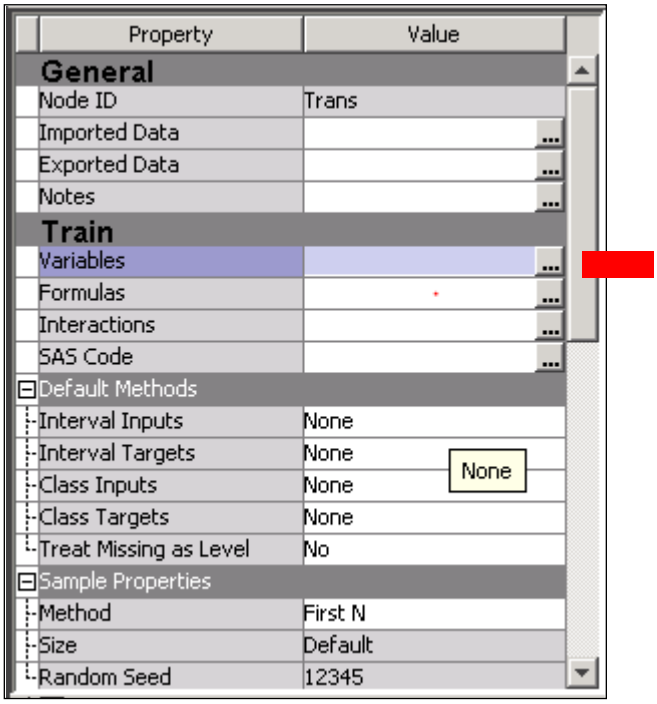
1. Select the **Modify** tab.

2. Drag the **Transform Variables** tool into the diagram workspace.

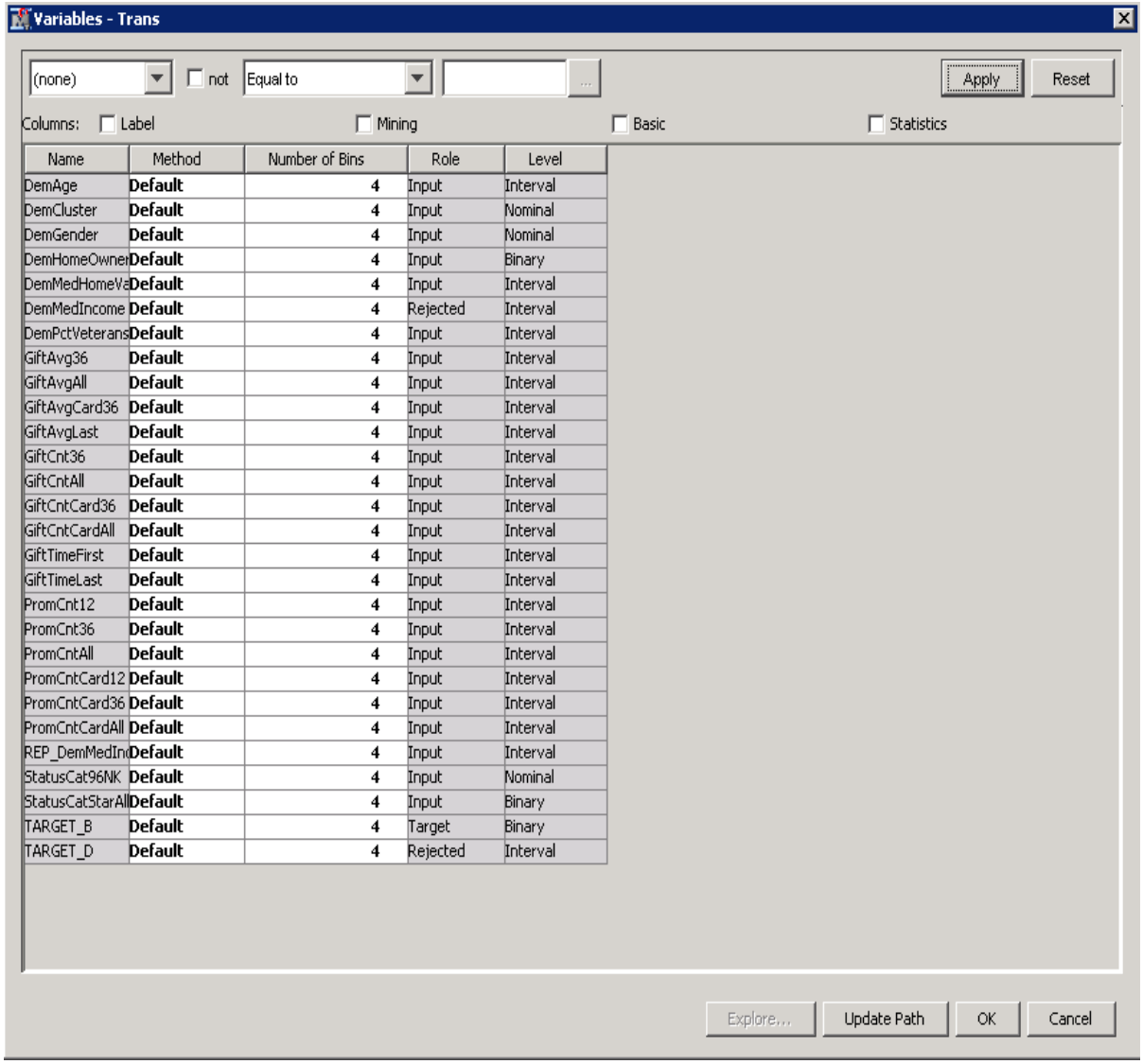
3. Connect the **Data Partition** node to the **Transform Variables** node.



4. Select **Variables** ⇨ ... for the property of the Transform Variables node.

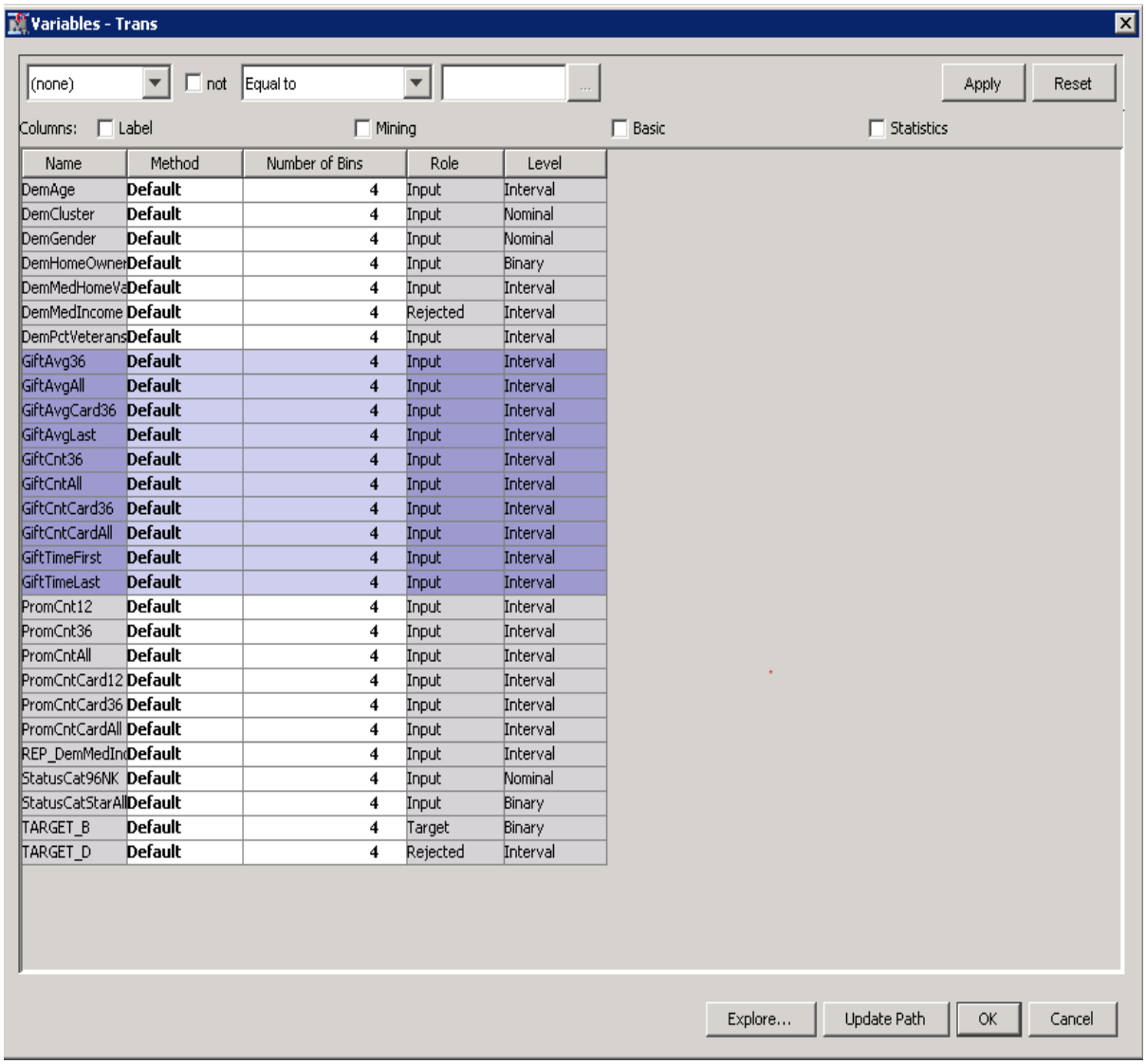


The Variables - Trans window appears.



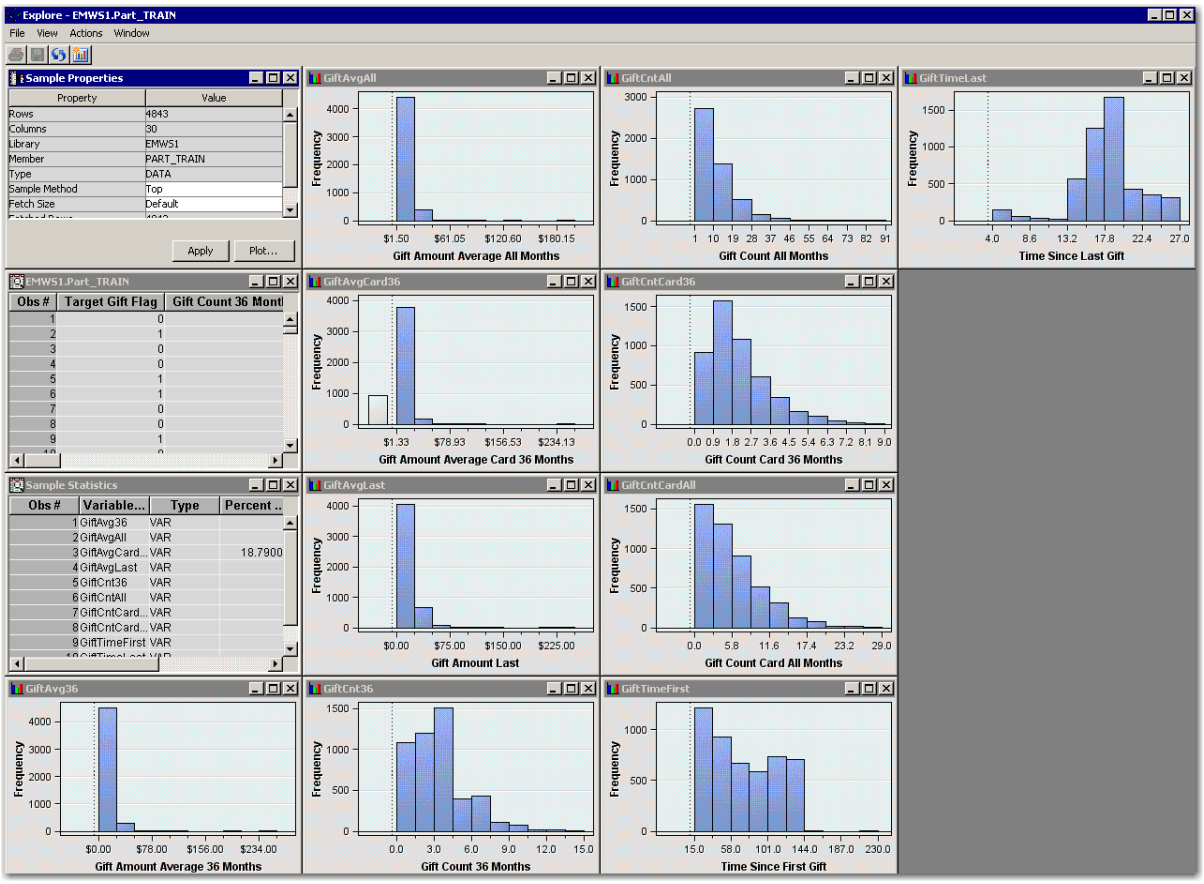
Name	Method	Number of Bins	Role	Level
DemAge	**Default**	4	Input	Interval
DemCluster	**Default**	4	Input	Nominal
DemGender	**Default**	4	Input	Nominal
DemHomeOwner	**Default**	4	Input	Binary
DemMedHomeVa	**Default**	4	Input	Interval
DemMedIncome	**Default**	4	Rejected	Interval
DemPctVeterans	**Default**	4	Input	Interval
GiftAvg36	**Default**	4	Input	Interval
GiftAvgAll	**Default**	4	Input	Interval
GiftAvgCard36	**Default**	4	Input	Interval
GiftAvgLast	**Default**	4	Input	Interval
GiftCnt36	**Default**	4	Input	Interval
GiftCntAll	**Default**	4	Input	Interval
GiftCntCard36	**Default**	4	Input	Interval
GiftCntCardAll	**Default**	4	Input	Interval
GiftTimeFirst	**Default**	4	Input	Interval
GiftTimeLast	**Default**	4	Input	Interval
PromCnt12	**Default**	4	Input	Interval
PromCnt36	**Default**	4	Input	Interval
PromCntAll	**Default**	4	Input	Interval
PromCntCard12	**Default**	4	Input	Interval
PromCntCard36	**Default**	4	Input	Interval
PromCntCardAll	**Default**	4	Input	Interval
REP_DemMedIn	**Default**	4	Input	Interval
StatusCat96NK	**Default**	4	Input	Nominal
StatusCatStarAll	**Default**	4	Input	Binary
TARGET_B	**Default**	4	Target	Binary
TARGET_D	**Default**	4	Rejected	Interval

5. Select all inputs with **Gift** in the name.



Name	Method	Number of Bins	Role	Level
DemAge	**Default**	4	Input	Interval
DemCluster	**Default**	4	Input	Nominal
DemGender	**Default**	4	Input	Nominal
DemHomeOwner	**Default**	4	Input	Binary
DemMedHomeVa	**Default**	4	Input	Interval
DemMedIncome	**Default**	4	Rejected	Interval
DemPctVeterans	**Default**	4	Input	Interval
GiftAvg36	**Default**	4	Input	Interval
GiftAvgAll	**Default**	4	Input	Interval
GiftAvgCard36	**Default**	4	Input	Interval
GiftAvgLast	**Default**	4	Input	Interval
GiftCnt36	**Default**	4	Input	Interval
GiftCntAll	**Default**	4	Input	Interval
GiftCntCard36	**Default**	4	Input	Interval
GiftCntCardAll	**Default**	4	Input	Interval
GiftTimeFirst	**Default**	4	Input	Interval
GiftTimeLast	**Default**	4	Input	Interval
PromCnt12	**Default**	4	Input	Interval
PromCnt36	**Default**	4	Input	Interval
PromCntAll	**Default**	4	Input	Interval
PromCntCard12	**Default**	4	Input	Interval
PromCntCard36	**Default**	4	Input	Interval
PromCntCardAll	**Default**	4	Input	Interval
REP_DemMedIn	**Default**	4	Input	Interval
StatusCat96NK	**Default**	4	Input	Nominal
StatusCatStarAll	**Default**	4	Input	Binary
TARGET_B	**Default**	4	Target	Binary
TARGET_D	**Default**	4	Rejected	Interval

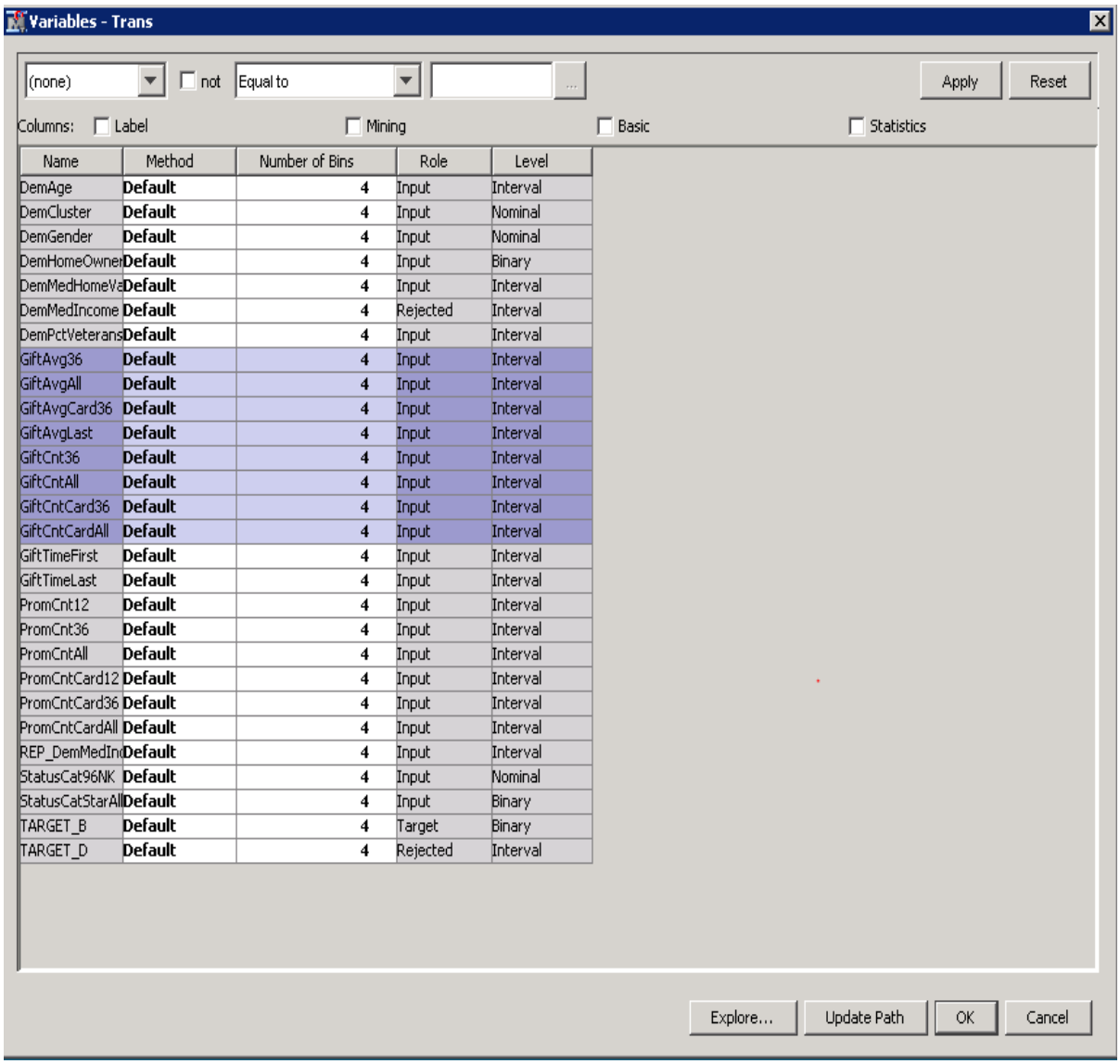
6. Select **Explore**. The Explore window appears.



The **GiftAvg** and **GiftCnt** inputs show some degree of skew in their distributions. The **GiftTime** inputs do not. To regularize the skewed distributions, use the log transformation. For these inputs, the order of magnitude of the underlying measure predicts the target rather than the measure itself.

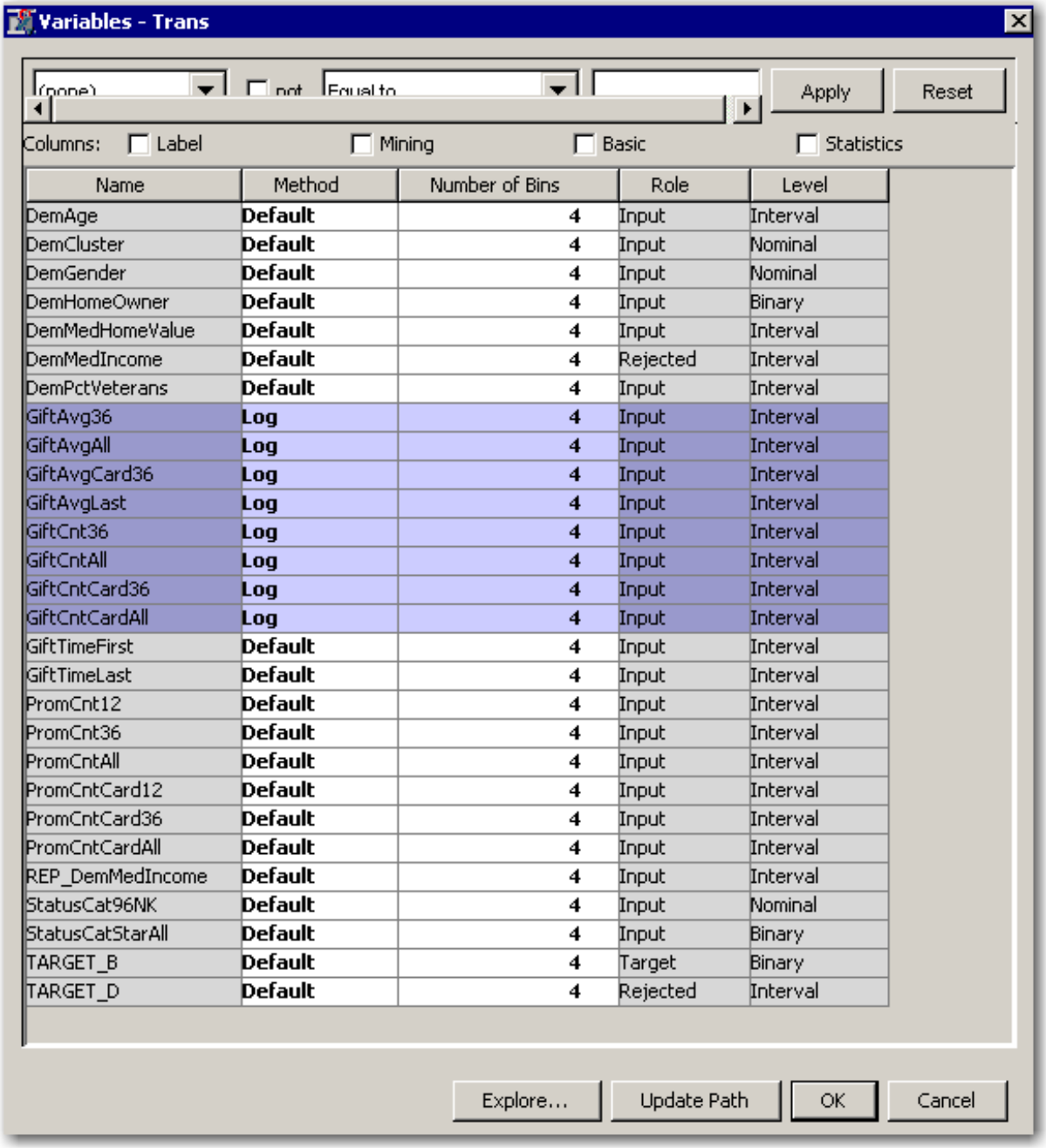
7. Close the Explore window.

8. Deselect the two inputs with **GiftTime** in their names.



Name	Method	Number of Bins	Role	Level
DemAge	**Default**	4	Input	Interval
DemCluster	**Default**	4	Input	Nominal
DemGender	**Default**	4	Input	Nominal
DemHomeOwner	**Default**	4	Input	Binary
DemMedHomeVa	**Default**	4	Input	Interval
DemMedIncome	**Default**	4	Rejected	Interval
DemPctVeterans	**Default**	4	Input	Interval
GiftAvg36	**Default**	4	Input	Interval
GiftAvgAll	**Default**	4	Input	Interval
GiftAvgCard36	**Default**	4	Input	Interval
GiftAvgLast	**Default**	4	Input	Interval
GiftCnt36	**Default**	4	Input	Interval
GiftCntAll	**Default**	4	Input	Interval
GiftCntCard36	**Default**	4	Input	Interval
GiftCntCardAll	**Default**	4	Input	Interval
GiftTimeFirst	**Default**	4	Input	Interval
GiftTimeLast	**Default**	4	Input	Interval
PromCnt12	**Default**	4	Input	Interval
PromCnt36	**Default**	4	Input	Interval
PromCntAll	**Default**	4	Input	Interval
PromCntCard12	**Default**	4	Input	Interval
PromCntCard36	**Default**	4	Input	Interval
PromCntCardAll	**Default**	4	Input	Interval
REP_DemMedInc	**Default**	4	Input	Interval
StatusCat96NK	**Default**	4	Input	Nominal
StatusCatStarAll	**Default**	4	Input	Binary
TARGET_B	**Default**	4	Target	Binary
TARGET_D	**Default**	4	Rejected	Interval

9. Select **Method** ⇨ **Log** for one of the remaining selected inputs. The selected method changes from **`Default`** to **`Log`** for the **GiftAvg** and **GiftCnt** inputs.



Name	Method	Number of Bins	Role	Level
DemAge	**Default**	4	Input	Interval
DemCluster	**Default**	4	Input	Nominal
DemGender	**Default**	4	Input	Nominal
DemHomeOwner	**Default**	4	Input	Binary
DemMedHomeValue	**Default**	4	Input	Interval
DemMedIncome	**Default**	4	Rejected	Interval
DemPctVeterans	**Default**	4	Input	Interval
GiftAvg36	**Log**	4	Input	Interval
GiftAvgAll	**Log**	4	Input	Interval
GiftAvgCard36	**Log**	4	Input	Interval
GiftAvgLast	**Log**	4	Input	Interval
GiftCnt36	**Log**	4	Input	Interval
GiftCntAll	**Log**	4	Input	Interval
GiftCntCard36	**Log**	4	Input	Interval
GiftCntCardAll	**Log**	4	Input	Interval
GiftTimeFirst	**Default**	4	Input	Interval
GiftTimeLast	**Default**	4	Input	Interval
PromCnt12	**Default**	4	Input	Interval
PromCnt36	**Default**	4	Input	Interval
PromCntAll	**Default**	4	Input	Interval
PromCntCard12	**Default**	4	Input	Interval
PromCntCard36	**Default**	4	Input	Interval
PromCntCardAll	**Default**	4	Input	Interval
REP_DemMedIncome	**Default**	4	Input	Interval
StatusCat96NK	**Default**	4	Input	Nominal
StatusCatStarAll	**Default**	4	Input	Binary
TARGET_B	**Default**	4	Target	Binary
TARGET_D	**Default**	4	Rejected	Interval

10. Select **OK** to close the Variables - Trans window.

11. Run the Transform Variables node and view the results.

12. Maximize the Output window and go to line 28.

```
              Input
Input Name    Role   Level      Name      Level       Formula

GiftAvg36    INPUT  INTERVAL  LOG_GiftAvg36    INTERVAL  log(GiftAvg36 + 1)
GiftAvgAll    INPUT  INTERVAL  LOG_GiftAvgAll    INTERVAL  log(GiftAvgAll + 1)
GiftAvgCard36  INPUT  INTERVAL  LOG_GiftAvgCard36  INTERVAL  log(GiftAvgCard36 + 1)
GiftAvgLast   INPUT  INTERVAL  LOG_GiftAvgLast   INTERVAL  log(GiftAvgLast + 1)
GiftCnt36    INPUT  INTERVAL  LOG_GiftCnt36    INTERVAL  log(GiftCnt36 + 1)
GiftCntAll    INPUT  INTERVAL  LOG_GiftCntAll    INTERVAL  log(GiftCntAll + 1)
GiftCntCard36  INPUT  INTERVAL  LOG_GiftCntCard36  INTERVAL  log(GiftCntCard36 + 1)
GiftCntCardAll  INPUT  INTERVAL  LOG_GiftCntCardAll  INTERVAL  log(GiftCntCardAll + 1)
```

Notice the Formula column. While a log transformation was specified, the actual transformation used was log(*input* + 1). This default action of the Transform Variables tool avoids problems with 0-values of the underlying inputs.

13. Close the Transform Variables - Results window.

Managing Missing Values (Using Imputation)

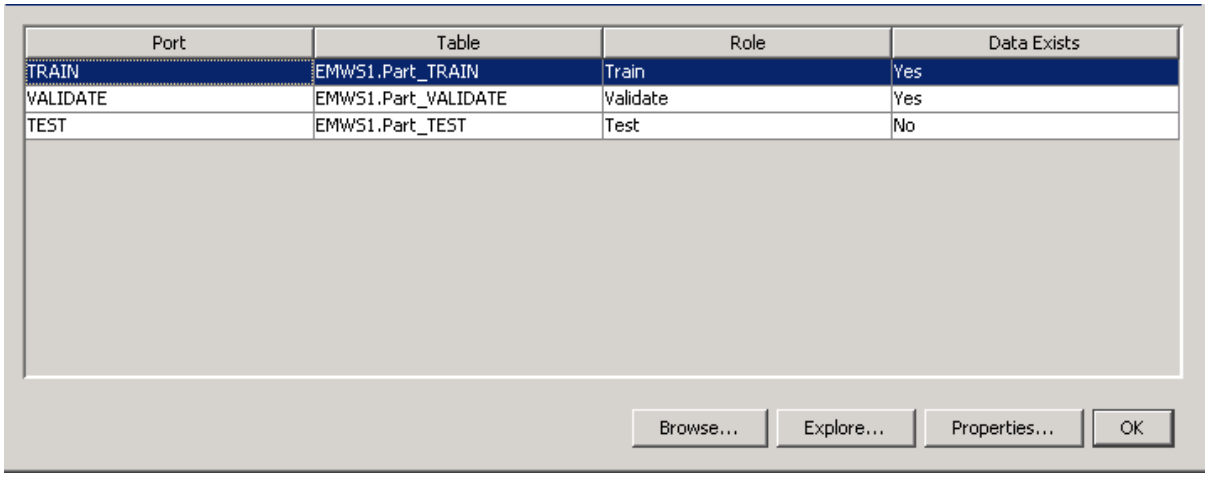
Data Assessment

As discussed above, regression requires that a case have a complete set of input values for both training and scoring. Follow these steps to examine the data status after the partition.

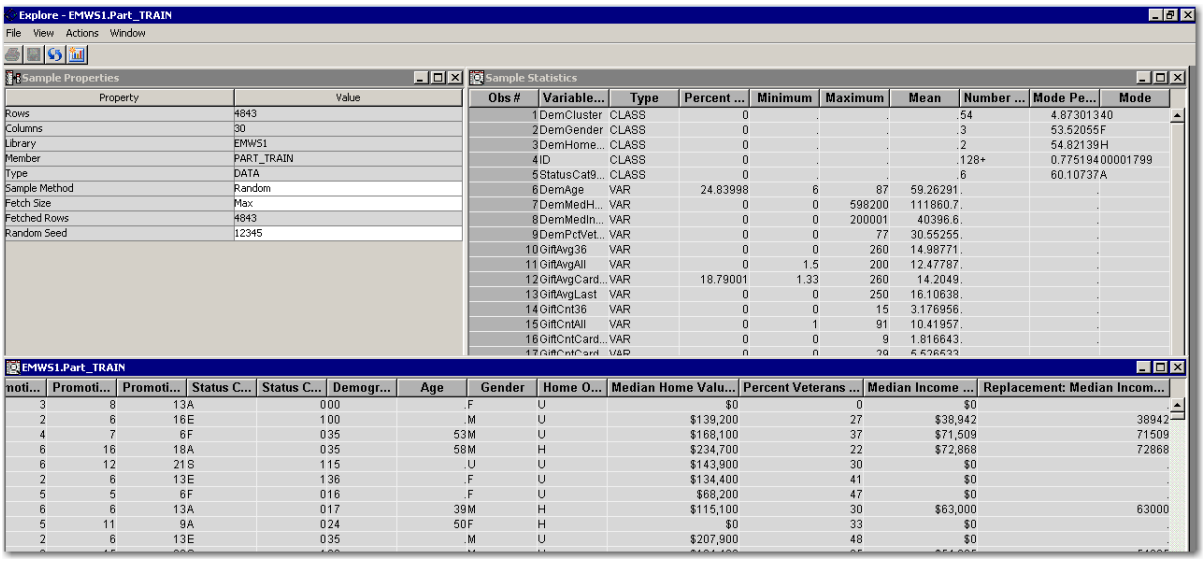
1. Select the **Transform Variables** node.

2. Select **Exported Data** ⇨ ... from the Transform Variables node property sheet.

3. Select the **TRAIN** data port and select **Explore**.



There are several inputs with a noticeable frequency of missing values, for example, **Age** and the replaced value of median income.



Explore - EMWS1.Part_TRAIN

File View Actions Window

Sample Properties

Property	Value
Rows	4843
Columns	30
Library	EMWS1
Member	PART_TRAIN
Type	DATA
Sample Method	Random
Fetch Size	Max
Fetched Rows	4843
Random Seed	12345

Sample Statistics

Obs #	Variable...	Type	Percent ...	Minimum	Maximum	Mean	Number ...	Mode Pe...	Mode
1	DemCluster	CLASS	0	.	.	.	.54	4.87301340	
2	DemGender	CLASS	0	.	.	.	.3	53.52055F	
3	DemHome...	CLASS	0	.	.	.	.2	54.82139H	
4	ID	CLASS	0	.	.	.	.128+	0.77519400001799	
5	StatusCat9...	CLASS	0	.	.	.	.6	60.10737A	
6	DemAge	VAR	24.83998	6	87	59.26291	.	.	
7	DemMedH...	VAR	0	0	598200	111860.7	.	.	
8	DemMedIn...	VAR	0	0	200001	40396.6	.	.	
9	DemPctVet...	VAR	0	0	77	30.55255	.	.	
10	GiftAvg36	VAR	0	0	260	14.98771	.	.	
11	GiftAvgAll	VAR	0	1.5	200	12.47787	.	.	
12	GiftAvgCard...	VAR	18.79001	1.33	260	14.2049	.	.	
13	GiftAvgLast	VAR	0	0	250	16.10638	.	.	
14	GiftCnt36	VAR	0	0	15	3.176956	.	.	
15	GiftCntAll	VAR	0	1	91	10.41957	.	.	
16	GiftCntCard...	VAR	0	0	9	1.816643	.	.	
17	GiftCntCard...	VAR	0	0	29	5.526533			

EMWS1.Part_TRAIN

noti...	Promoti...	Promoti...	Status C...	Status C...	Demogr...	Age	Gender	Home O...	Median Home Valu...	Percent Veterans ...	Median Income ...	Replacement: Median Incom...
3	8	13A		000		.	F	U	$0	0	$0	.
2	6	16E		100		.	M	U	$139,200	27	$38,942	38942
4	7	6F		035	53	M	U	U	$168,100	37	$71,509	71509
6	16	18A		035	58	M	H	U	$234,700	22	$72,868	72868
6	12	21S		115		.	U	U	$143,900	30	$0	.
2	6	13E		136		.	F	U	$134,400	41	$0	.
5	5	6F		016		.	F	U	$68,200	47	$0	.
6	6	13A		017	39	M	H	U	$115,100	30	$63,000	63000
5	11	9A		024	50	F	H	U	$0	33	$0	.
2	6	13E		035		.	M	U	$207,900	48	$0	.

Median Home Valu...	Percent Veterans ...	Median Income ...	Replacement: Median Incom...
$0	0	$0	.
$139,200	27	$38,942	38942
$168,100	37	$71,509	71509
$234,700	22	$72,868	72868
$143,900	30	$0	.
$134,400	41	$0	.
$68,200	47	$0	.
$115,100	30	$63,000	63000
$0	33	$0	.
$207,900	48	$0	.
$104,400	35	$54,385	54385

There are several ways to proceed:

- ***Do nothing.*** If there are very few cases with missing values, this is a viable option. The difficulty with this approach comes when the model must predict a new case that contains a missing value. Omitting the missing term from the parametric equation usually produces an extremely biased prediction.

- ***Impute*** a synthetic value for the missing value. For example, if an interval input contains a missing value, replace the missing value with the mean of the nonmissing values for the input. This eliminates the incomplete case problem but modifies the input's distribution. This can bias the model predictions.

  Making the missing value imputation process part of the modeling process allays the modified distribution concern. Any modifications made to the training data are also made to the validation data and the remainder of the modeling population. A model trained with the modified training data will not be biased if the same modifications are made to any other data set that the model might encounter (and the data has a similar pattern of missing values).

- Create a ***missing indicator*** for each input in the data set. Cases often contain missing values for a reason. If the reason for the missing value is in some way related to the target variable, useful predictive information is lost.

  The missing indicator is 1 when the corresponding input is missing and 0 otherwise. Each missing indicator becomes an input to the model. This enables modeling of the association between the target and a missing value on an input.

4. Close the Explore and Exported Data windows.

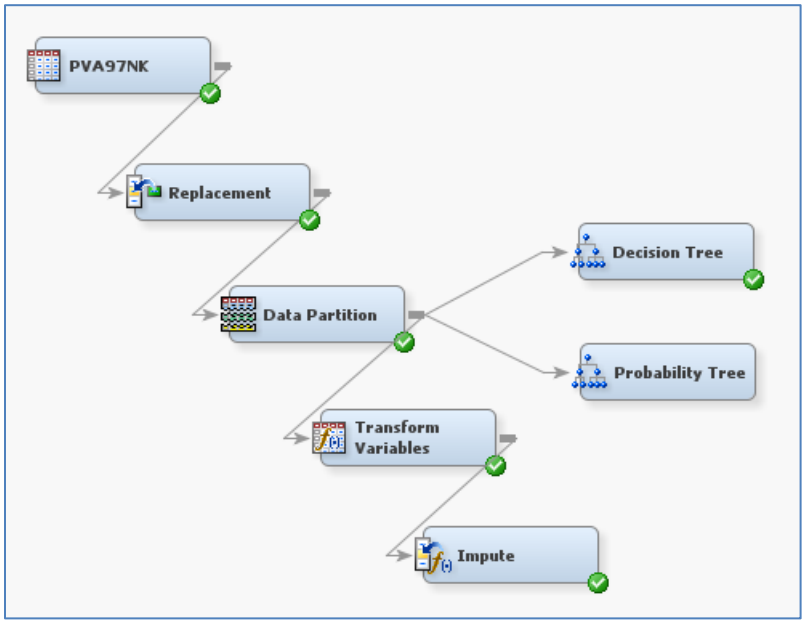
Imputation

To address missing values in the **PVA97NK** data set, use the following steps to impute synthetic data values and create missing value indicators:

1. Select the **Modify** tab.

2. Drag an **Impute** tool into the diagram workspace.

3. Connect the **Transform Variables** node to the **Impute** node.



4. Select the **Impute** node and examine the Properties panel.

Property	Value
General	
Node ID	Impt
Imported Data	
Exported Data	
Notes	
Train	
Variables	
Non Missing Variables	No
Missing Cutoff	50.0
Class Variables	
Default Input Method	Count
Default Target Method	None
Normalize Values	Yes
Interval Variables	
Default Input Method	Mean
Default Target Method	None
Default Constant Value	
Default Character Value	
Default Number Value	.
Method Options	
Random Seed	12345
Tuning Parameters	
Tree Imputation	
Score	
Hide Original Variables	Yes
Indicator Variables	
Type	None
Source	Imputed Variables
Role	Rejected

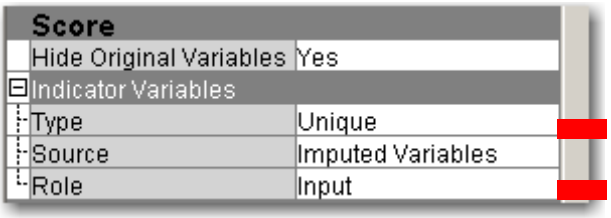
The defaults of the Impute node are as follows:

- For interval inputs, replace any missing values with the mean of the nonmissing values.
- For categorical inputs, replace any missing values with the most frequent category.

With these settings, each input with missing values generates a new input. The new input named IMP_*original_input_name* will have missing values replaced by a synthetic value and nonmissing values copied from the original input.

Missing Indicators

Use the steps below to create missing value indicators. The settings for missing value indicators are found in the Score property group.



Score	
Hide Original Variables	Yes
Indicator Variables	
Type	Unique
Source	Imputed Variables
Role	Input

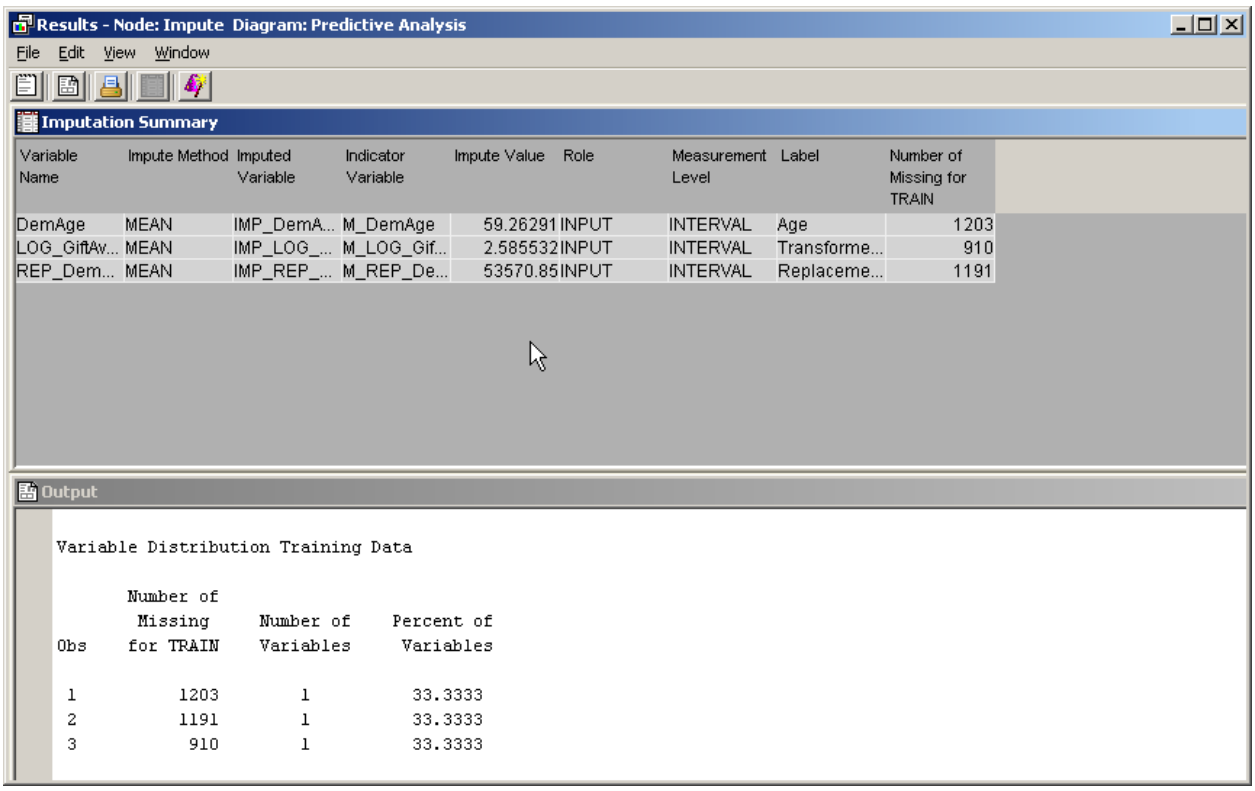
1. Select **Indicator Variables** ⇨ **Type** ⇨ **Unique**.

2. Select **Indicator Variables** ⇨ **Role** ⇨ **Input**.

With these settings, new inputs named M_*original_input_name* are added to the training data to indicate the synthetic data values.

Imputation Results

Run the Impute node and review the Results window. Three inputs had missing values.



Variable Name	Impute Method	Imputed Variable	Indicator Variable	Impute Value	Role	Measurement Level	Label	Number of Missing for TRAIN
DemAge	MEAN	IMP_DemA...	M_DemAge	59.26291	INPUT	INTERVAL	Age	1203
LOG_GiftAv...	MEAN	IMP_LOG_...	M_LOG_Gif...	2.585532	INPUT	INTERVAL	Transforme...	910
REP_Dem...	MEAN	IMP_REP_...	M_REP_De...	53570.85	INPUT	INTERVAL	Replaceme...	1191

Variable Distribution Training Data

Obs	Number of Missing for TRAIN	Number of Variables	Percent of Variables
1	1203	1	33.3333
2	1191	1	33.3333
3	910	1	33.3333

Running a Regression Node

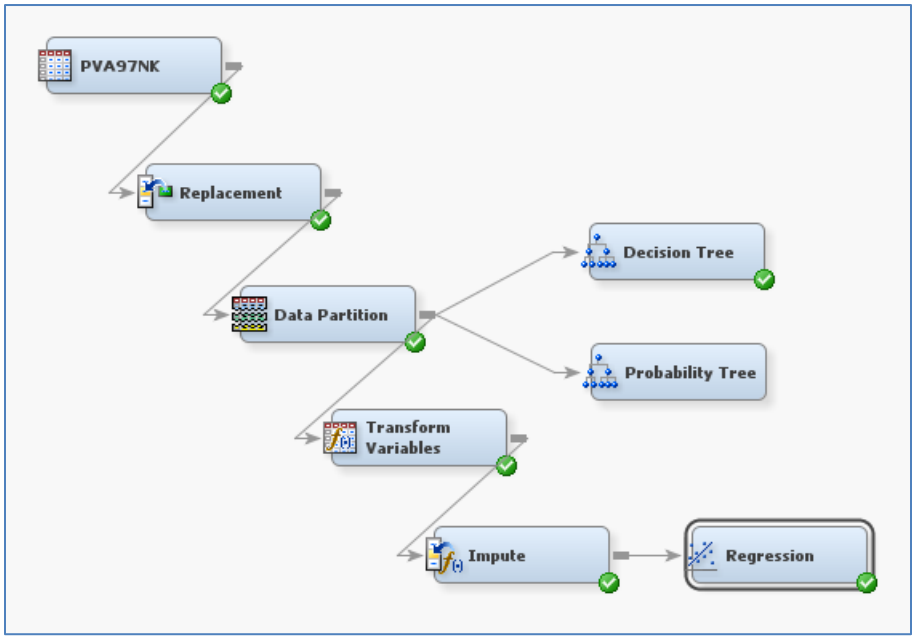
There are several tools in SAS Enterprise Miner to fit regression or regression-like models. By far, the most commonly used (and, arguably, the most useful) is the *Regression tool*.

Use the following steps to build a simple regression model.

1. Select the **Model** tab.

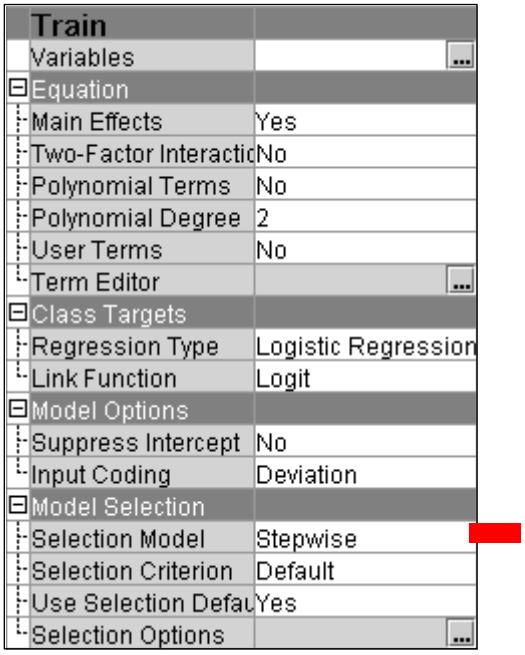
2. Drag a **Regression** tool into the diagram workspace.

3. Connect the **Impute** node to the **Regression** node.



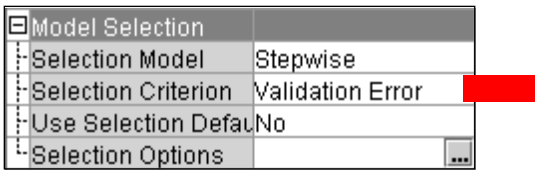
The Regression node can create several types of regression models, including linear and logistic. The type of default regression type is determined by the target's measurement level.

4. Select **Selection Model** ⇨ **Stepwise** on the Regression node property sheet.



The Regression node is now configured to use stepwise selection to choose inputs for the model.

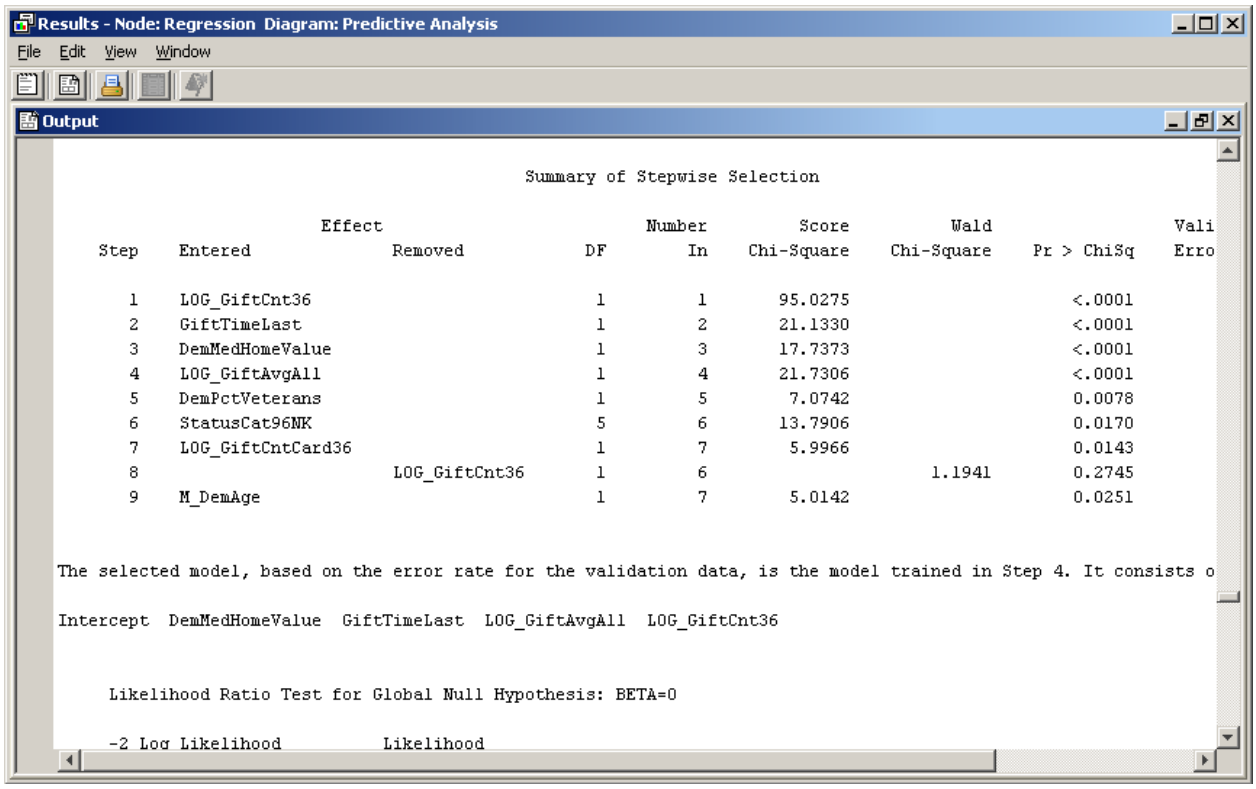
5. Suppose that your predictions are estimates (or rankings). And so, select **Selection Criterion** ⇨ **Validation Error**.



6. Run the **Regression** node and view the results.

7. Maximize the **Output** window.

8. Use CRTL-F to go to “Summary of Stepwise Selection”.



Summary of Stepwise Selection

Step	Effect Entered	Effect Removed	DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq	Vali Erro
1	LOG_GiftCnt36		1	1	95.0275		<.0001	
2	GiftTimeLast		1	2	21.1330		<.0001	
3	DemMedHomeValue		1	3	17.7373		<.0001	
4	LOG_GiftAvgAll		1	4	21.7306		<.0001	
5	DemPctVeterans		1	5	7.0742		0.0078	
6	StatusCat96NK		5	6	13.7906		0.0170	
7	LOG_GiftCntCard36		1	7	5.9966		0.0143	
8		LOG_GiftCnt36	1	6		1.1941	0.2745	
9	M_DemAge		1	7	5.0142		0.0251	

The selected model, based on the error rate for the validation data, is the model trained in Step 4. It consists o

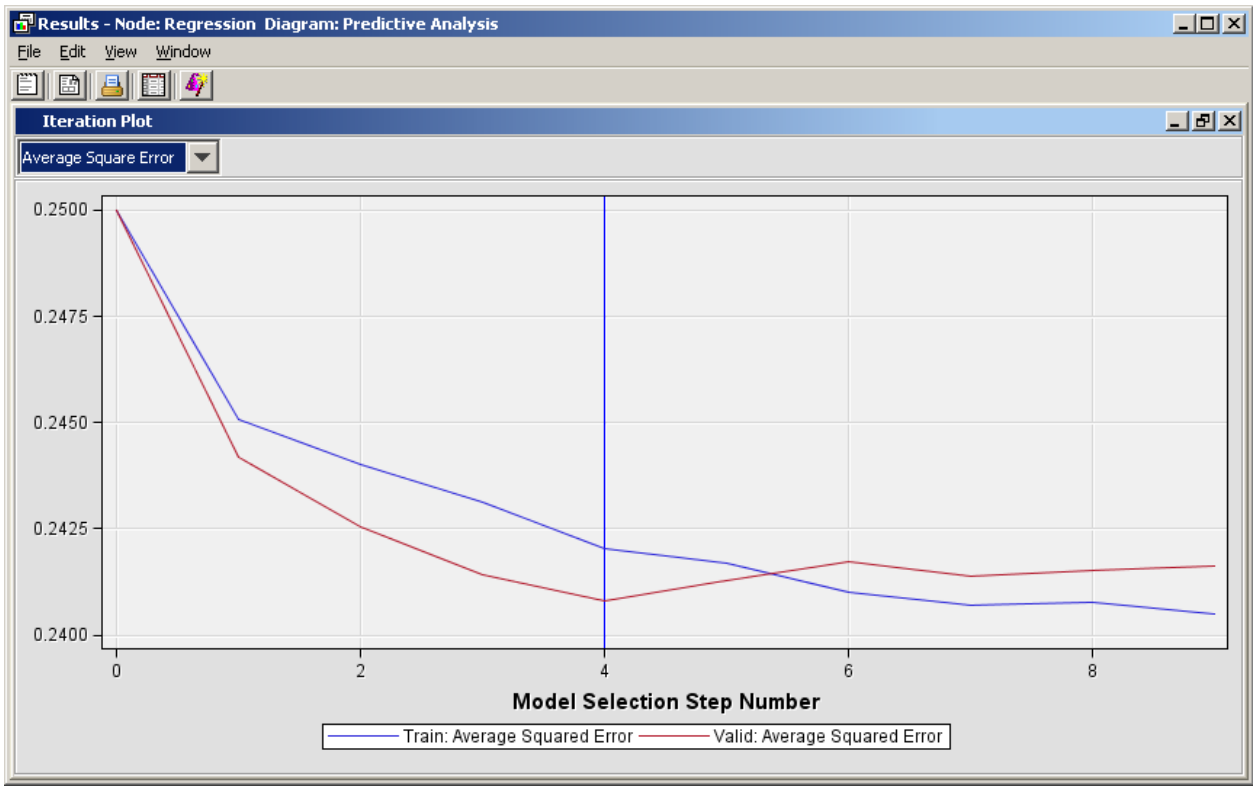
Intercept DemMedHomeValue GiftTimeLast LOG_GiftAvgAll LOG_GiftCnt36

Likelihood Ratio Test for Global Null Hypothesis: BETA=0

-2 Log Likelihood Likelihood

The stepwise selection process took 9 steps, and the selected model came from Step 4. Notice that half of the selected inputs are log transformations of the original gift variables.

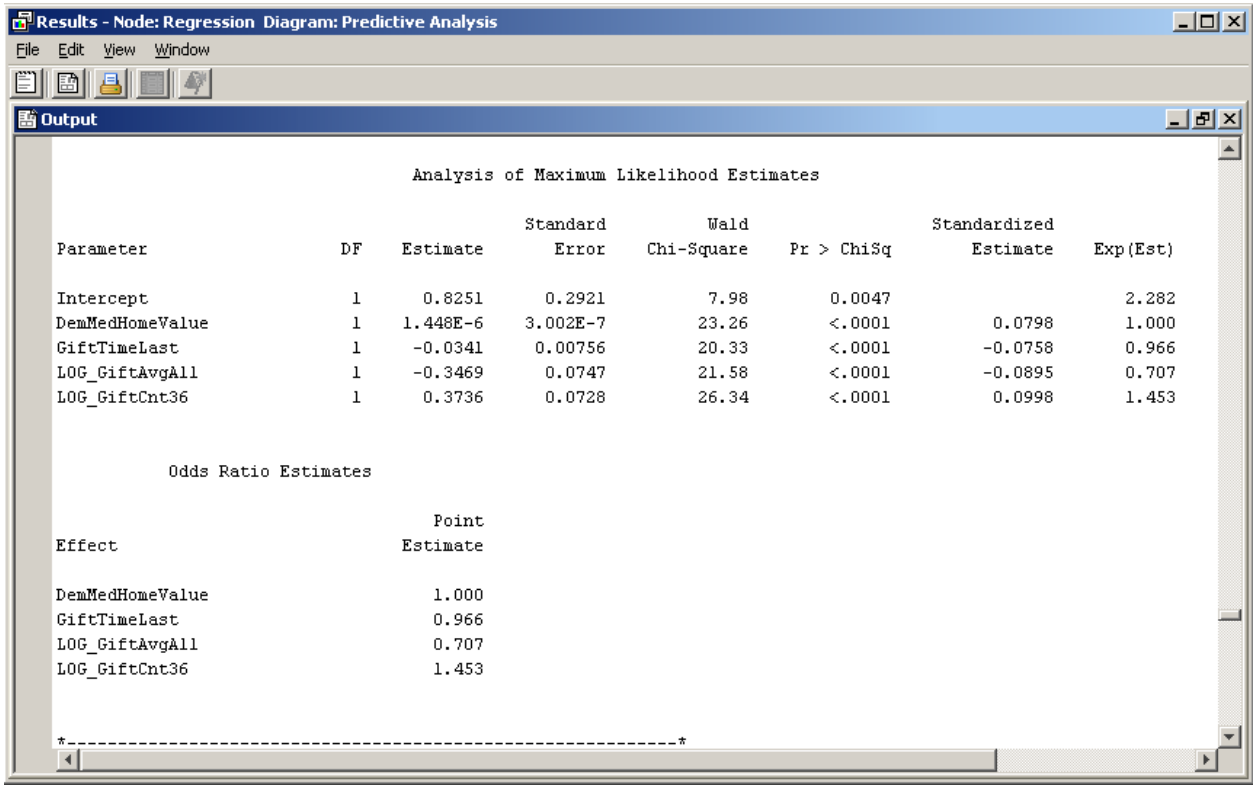
9. Select **View** ⇨ **Model** ⇨ **Iteration Plot**.



The selected model (based on minimum error) occurs in Step 4.

The following demonstrates how to interpret a model using odds ratios:

10. In the Output window, find the tables following the summary of the stepwise selection results.



Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq	Standardized Estimate	Exp(Est)
Intercept	1	0.8251	0.2921	7.98	0.0047		2.282
DemMedHomeValue	1	1.448E-6	3.002E-7	23.26	<.0001	0.0798	1.000
GiftTimeLast	1	-0.0341	0.00756	20.33	<.0001	-0.0758	0.966
LOG_GiftAvgAll	1	-0.3469	0.0747	21.58	<.0001	-0.0895	0.707
LOG_GiftCnt36	1	0.3736	0.0728	26.34	<.0001	0.0998	1.453

Odds Ratio Estimates

Effect	Point Estimate
DemMedHomeValue	1.000
GiftTimeLast	0.966
LOG_GiftAvgAll	0.707
LOG_GiftCnt36	1.453

For **LOG_GiftCnt36**, the odds ratio estimate equals 1.453. This means that for each unit increase in the log of the number of donations in the past 36 months, the odds of donation during the 97NK campaign change by a factor of 1.453, a 45.3% increase.

For **LOG_GiftAvgAll**, the odds ratio estimate equals 0.707. This means that for each unit increase in the log of the average dollar donated in the past, the odds of donation during the 97NK campaign change by a factor of 0.707, a 29.3% decrease.

For **GiftTimeLast**, the odds ratio estimate equals 0.966. This means that for each additional month in how long ago the last donation was made, the odds of donation during the 97NK campaign change by a factor of 0.966, a 3.4% decrease.

The unusual value of 1.000 for the **DemMedHomeValue** odds ratio has a simple explanation. Unit (that is, single dollar) changes in home value do not change the odds of response by an amount captured in three significant digits. To obtain a more meaningful value for this input's effect on response odds, you can multiply the parameter estimate by 1000 and exponentiate the result. You then have the change in response odds based on 1000 dollar changes in median home value. Equivalently, you could use the Transform Variables node to replace **DemMedHomeValue** with **DemMedHmVal1000=DemMedHomeValue/1000**, and a unit increase on that new input would represent a $1000 increase in the **DemMedHomeValue**.

11. Close the Results window.